Data Collection & Preprocessing

(Natural Language Processing)

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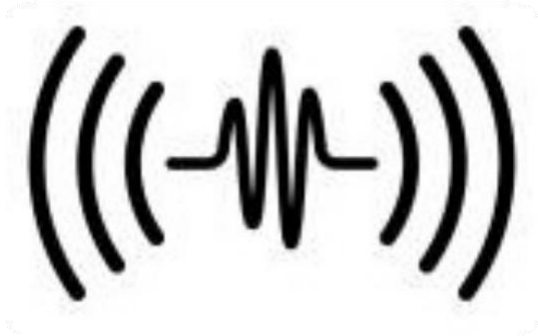
DATA
DATA
COLLECTION

Data collection facts

- Data collection, cleaning and preparation are estimated to take 70% of any data science task
- Quality and diversity of data play a significant role
- Sources of data a disparate (very different one from another)
- Defining clear objectives is essential for guiding the data collection process effectively

Potential Data Sources

- Data sources impact the quality, diversity, and applicability of the collected data
- Identify the kind of data you want to collect?
  - Text?
  - Numerical?
  - Speech?



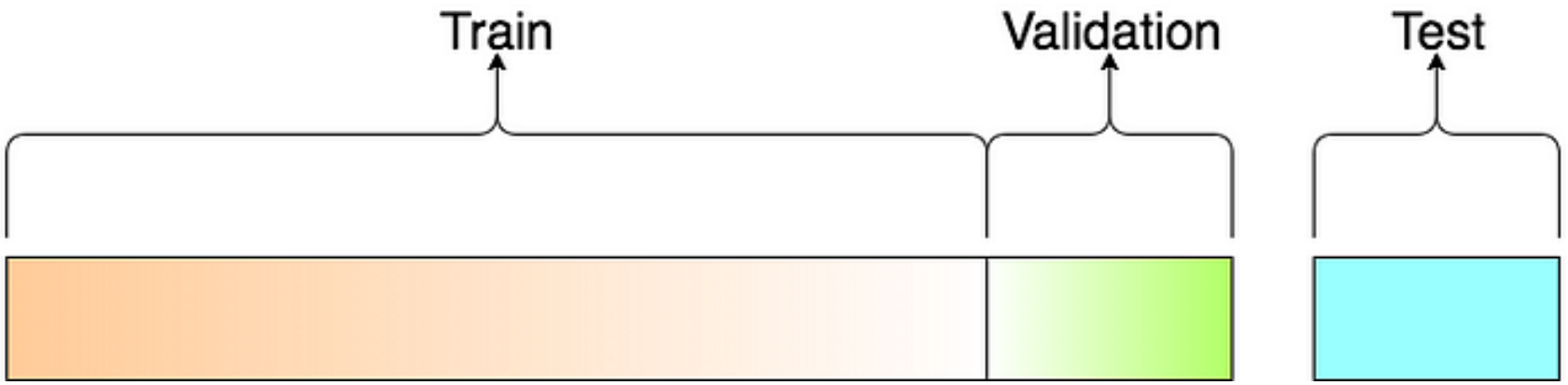
First Things First: Look at Your Data

REMEMBER

- We cannot use the data we used to build the model to evaluate it since the developed model always remembers the whole training dataset.
- Target is to generalize the model (Check whether the model performs well on new data)
- Split Dataset:
  - Training
  - Validation
  - Testing

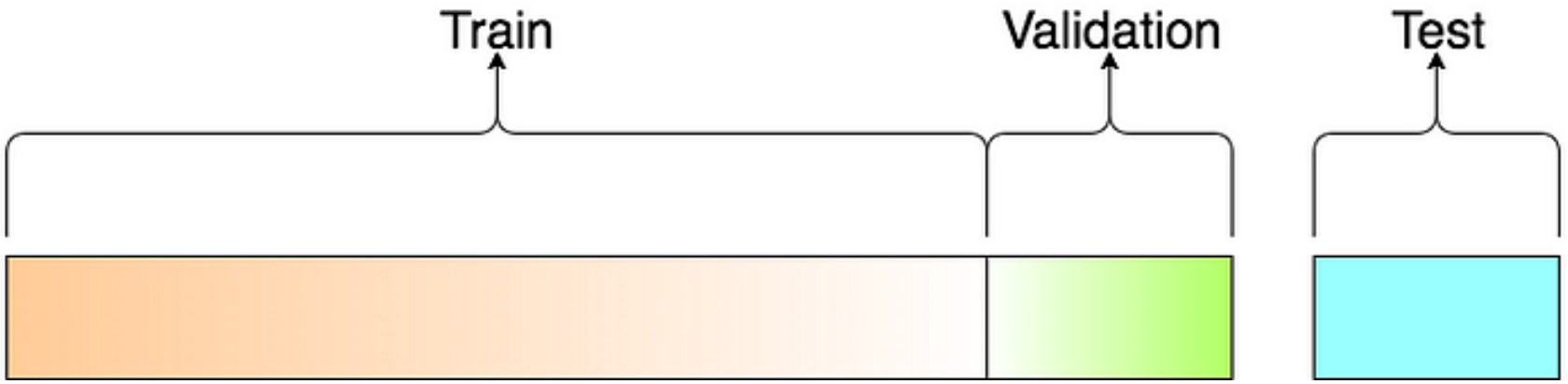
Training Dataset

- One part of the data is used to build our machine learning model
- The model *sees* and *learns* from this data.



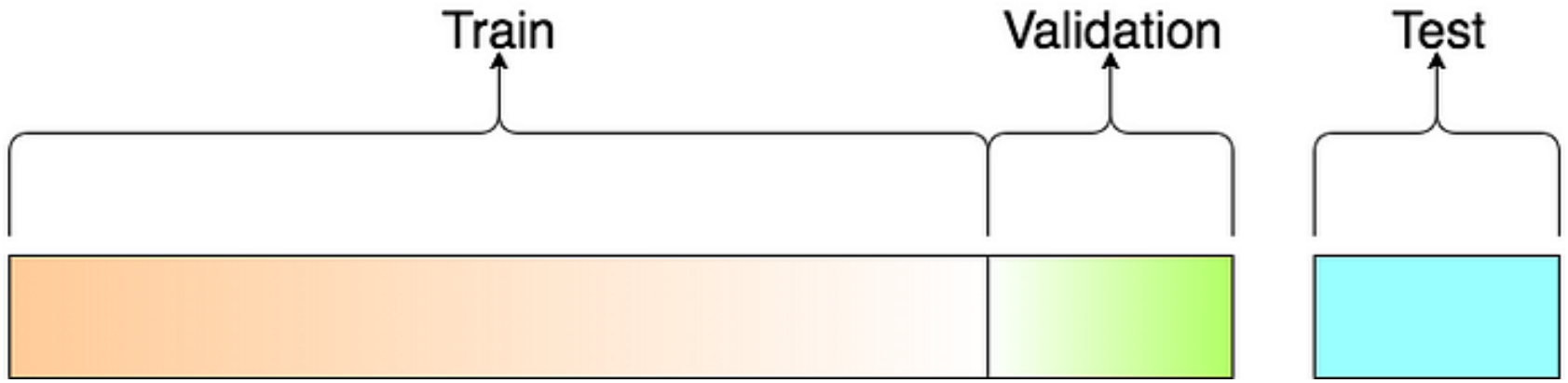
Validation Dataset

- The sample of data used to provide an unbiased evaluation of a model fit on the training dataset while tuning model hyperparameters.
- Use this data to fine-tune the model hyperparameters.
- The model occasionally sees this data, but never does it “Learn” from this.



Test Dataset

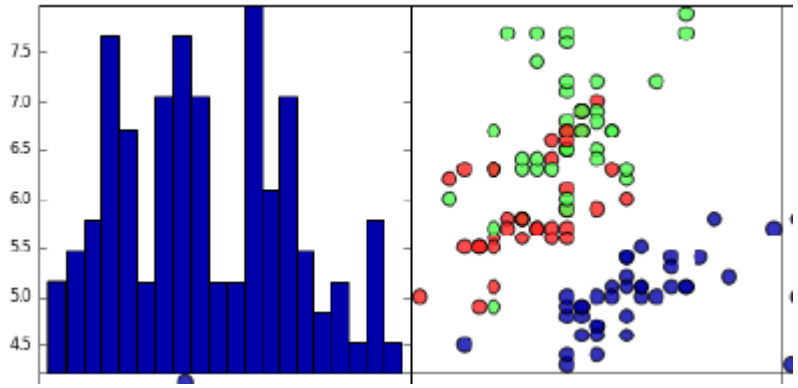
- Used to assess how well the model works.
- Used to provide an unbiased evaluation of a final model fit on the training dataset
- Only used once a model is completely trained(using the train and validation sets)



First Things First: Look at Your Data

REMEMBER:

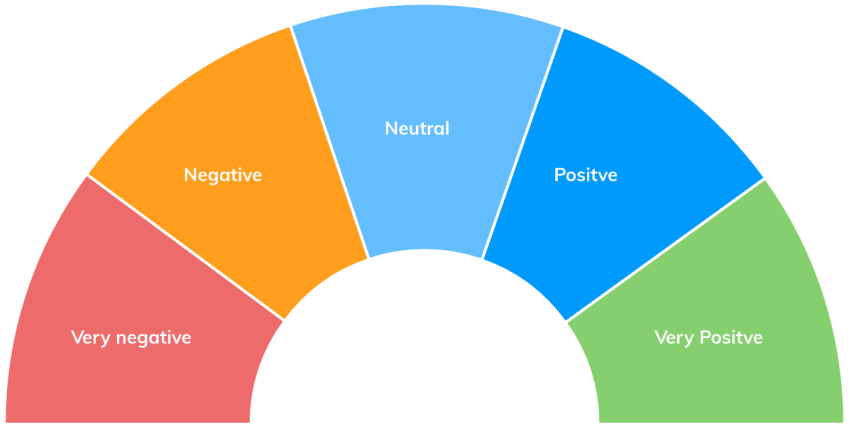
- Before building a machine learning model it is often a good idea to inspect the data, to see if the task is easily solvable without machine learning, or if the desired information might not be contained in the data.
- One of the best ways to inspect data is to visualize it.
  - Scatter Plots
  - Box and Whisker Plot for Large Data
  - Pie and Donut Charts



First Things First: Look at Your Data

Text Visualization/Analysis:

- Word Clouds
- Topic classification
- Sentiment Analysis/Aspect Based Sentiment Analysis



What kind of Data?

How much Data?

How to collect a Balanced Dataset?

Study the Dataset and Identify Features

How to divide Training, Validation and Testing Data?

What are the preprocessing steps required?

Features of a Dataset

- garbage in → garbage out
- System will only be capable of learning if the training data contains enough relevant features and not too many irrelevant ones.
- A critical part of the success of a machine learning project is coming up with a good set of features to train on. This process, called *feature engineering*
  - Feature selection
  - Feature extraction
  - Creating new features by gathering new data

Potential Data Sources

- Identify reliable data sources for the data you want to collect
  - Look for industry-accepted or professional data sources in your domain
  - Look for highly cited sources
  - Make a list of all possible sources & weigh the pros and cons
    - Services offered by data providers
    - Payment plan, if any
    - Frequency of update
    - Attributes of dataset
- Identify how the data might be collected legally from these sources

Potential Data Sources

Websites and Online Platforms:

- Websites, blogs, news articles, social media platforms, discussion forums, online communities.
- Example: Gathering tweets for sentiment analysis or news articles for summarization.

Text Corpora and Datasets:

- Curated datasets created for specific tasks, such as sentiment analysis, machine translation, question answering.
- Example: IMDb dataset for sentiment analysis, WMT translation datasets.

Potential Data Sources

Domain-Specific Data:

- Data from a particular domain, industry, or field (e.g., medical journals, legal documents).
- Example: Medical research articles for medical NLP tasks.

User-Generated Content:

- Content generated by users, such as customer reviews, product feedback, and online comments.
- Example: Amazon product reviews for sentiment analysis or user opinions.

Scientific Journals and Publications:

- Academic and research papers, publications in specific domains.
- Example: Academic papers for building domain-specific language models.

Potential Data Sources

- Factors to Consider:
  - Relevance: Ensure the data source aligns with the project's objectives and intended tasks.
  - Quality: Choose sources with credible, accurate, and well-written content.
  - Quantity: Know the amount of data needed.
  - Format: Data should be in a format easy to work with.
  - Diversity: Opt for sources that provide a diverse range of language styles, topics, and perspectives.
  - Accessibility: Consider data accessibility, licensing, and terms of use.

Data Collection Methods

Using APIs:

- Utilizing APIs provided by various platforms to fetch structured data.
- **Sources**: Twitter API, Reddit API, News APIs (like NewsAPI), Google Books API.
- **Applications**: Collecting social media posts, news articles, and other structured text data.

Scraping the Web:

- Extracting text data from websites using automated scripts.
- **Tools**: Beautiful Soup, Scrapy, Selenium.
- **Applications**: Building large corpora from news websites, forums, blogs, and social media.

Data Collection Methods

Public Datasets:

- Using datasets that have been pre-collected and made available by researchers or organizations.
- **Examples**:
  - **Text**: The Penn Treebank, WikiText, Project Gutenberg.
  - **Speech**: LibriSpeech, Common Voice.
- **Applications**: Training models on well-established benchmarks.

Crowdsourcing:

- Gathering data by engaging human participants through platforms like Amazon Mechanical Turk.
- **Tools**: Amazon Mechanical Turk, Prolific, Figure Eight (formerly CrowdFlower).
- **Applications**: Annotating text for sentiment analysis, collecting paraphrases, creating question-answer pairs.

Data Collection Methods

Manual Data Entry:

- Directly inputting data by researchers or hired annotators.
- **Applications**: Creating gold-standard datasets, fine-tuning specific tasks where precision is crucial, speech transcription.

Sensor Data and IoT:

- Collecting speech data through devices like smart speakers.
- **Tools**: Home assistants (Amazon Echo, Google Home).
- **Applications**: Building voice recognition systems, contextual NLP tasks in smart environments.

Data Collection Methods

Document Scanning and OCR:

- Digitizing printed text using Optical Character Recognition (OCR).
- **Tools**: Tesseract, Adobe Acrobat OCR.
- **Applications**: Creating digital versions of books, historical documents, legal papers.

Surveys and Interviews:

•Collecting structured and unstructured text through direct interactions.

•**Applications**: Gathering user opinions, conducting sentiment analysis on feedback.

Speech Data Collection:

Call Centers and Customer Service Interactions

•Recording conversations between customers and service agents.
•Analyzing real-world dialogue, improving customer service automation, sentiment analysis.

In-Lab Recording

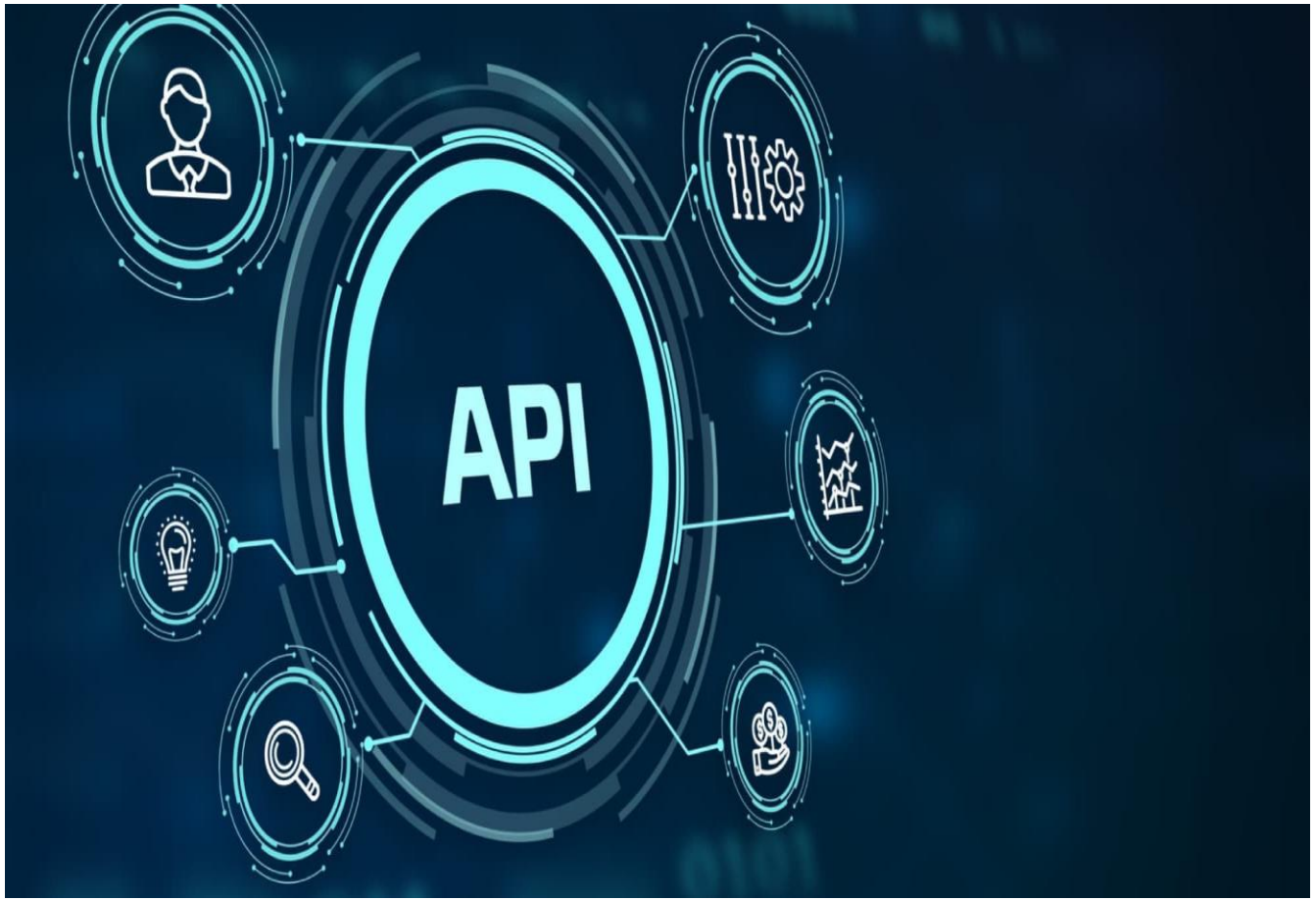
•Controlled environment recordings using high-quality equipment.
•Creating high-quality datasets, minimizing background noise.

Interview and Survey Recordings

•Recording responses from structured interviews or surveys.
•Collecting specific types of speech data, such as opinions, narratives, and structured responses.

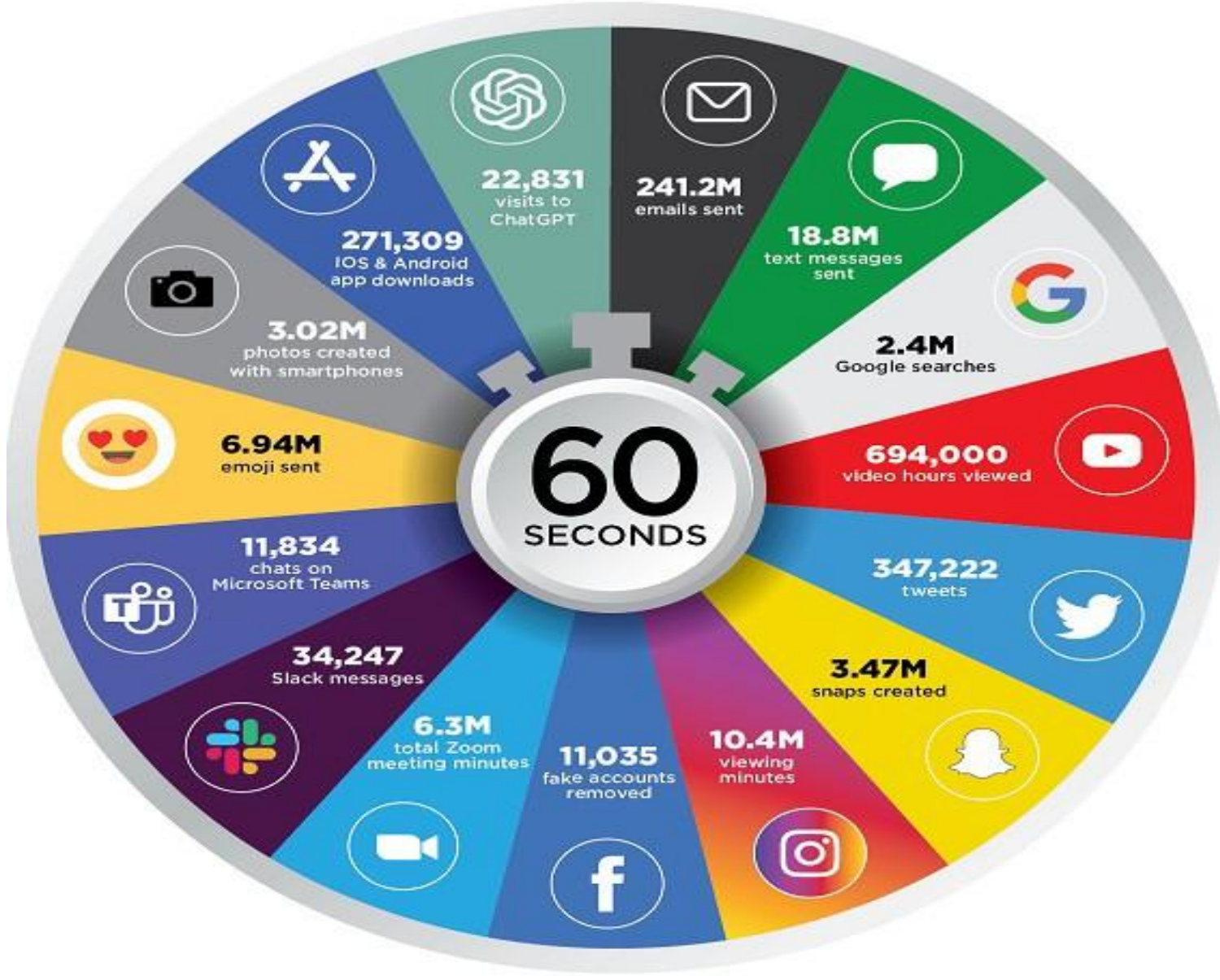
Broadcast Media

•Extracting speech from TV shows, radio, podcasts, and YouTube videos.
•Building large corpora, analyzing different speaking styles and contexts.



Data Collection through APIs

THE INTERNET IN 2023 EVERY MINUTE



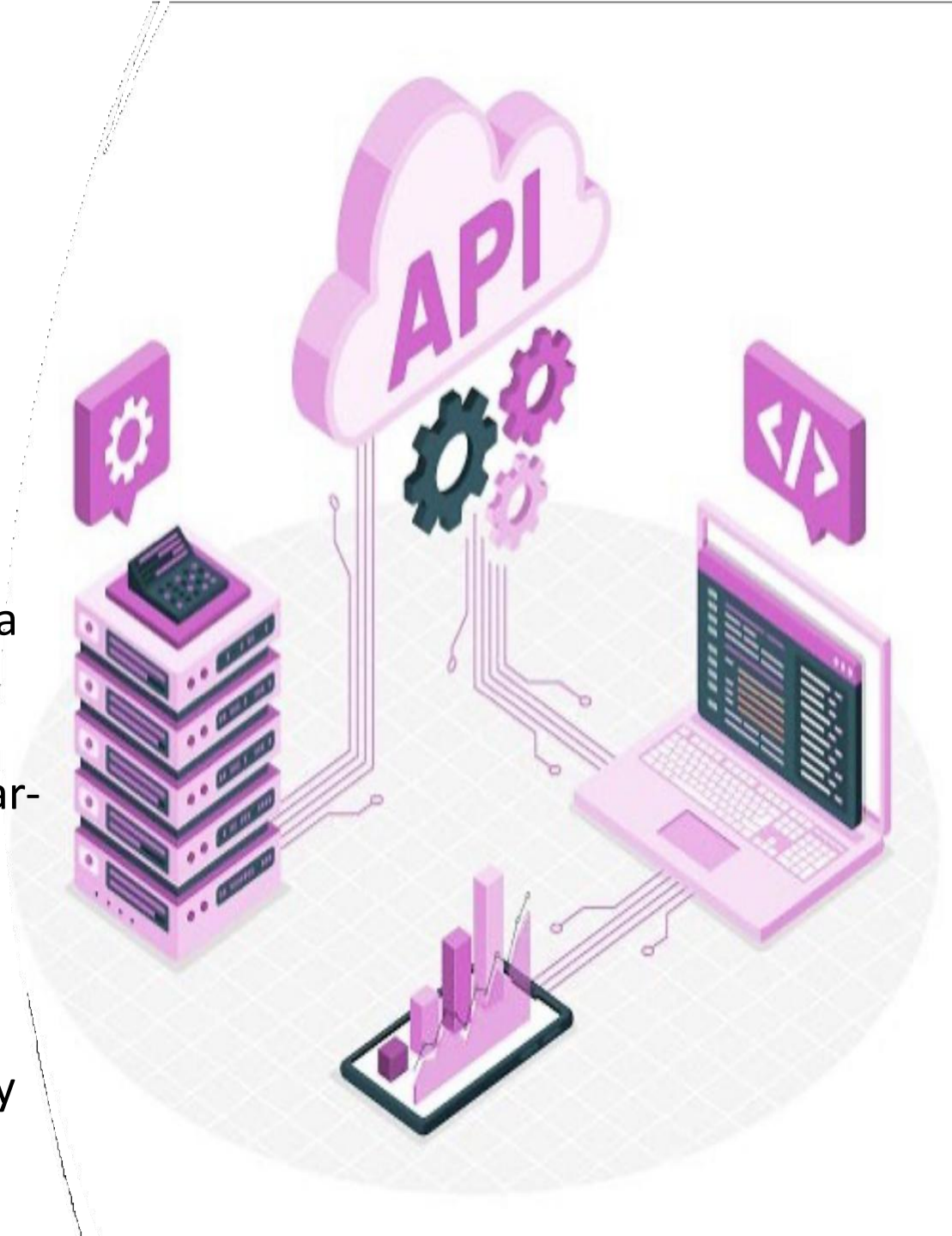
Created by: eDiscovery Today & LTMG

Data Collection through APIs

- What is an API?
  - An API is a set of rules and protocols that allow different software applications to communicate with each other.
  - APIs enable developers to request specific data or perform certain actions from remote servers, usually in a structured format like JSON or XML.

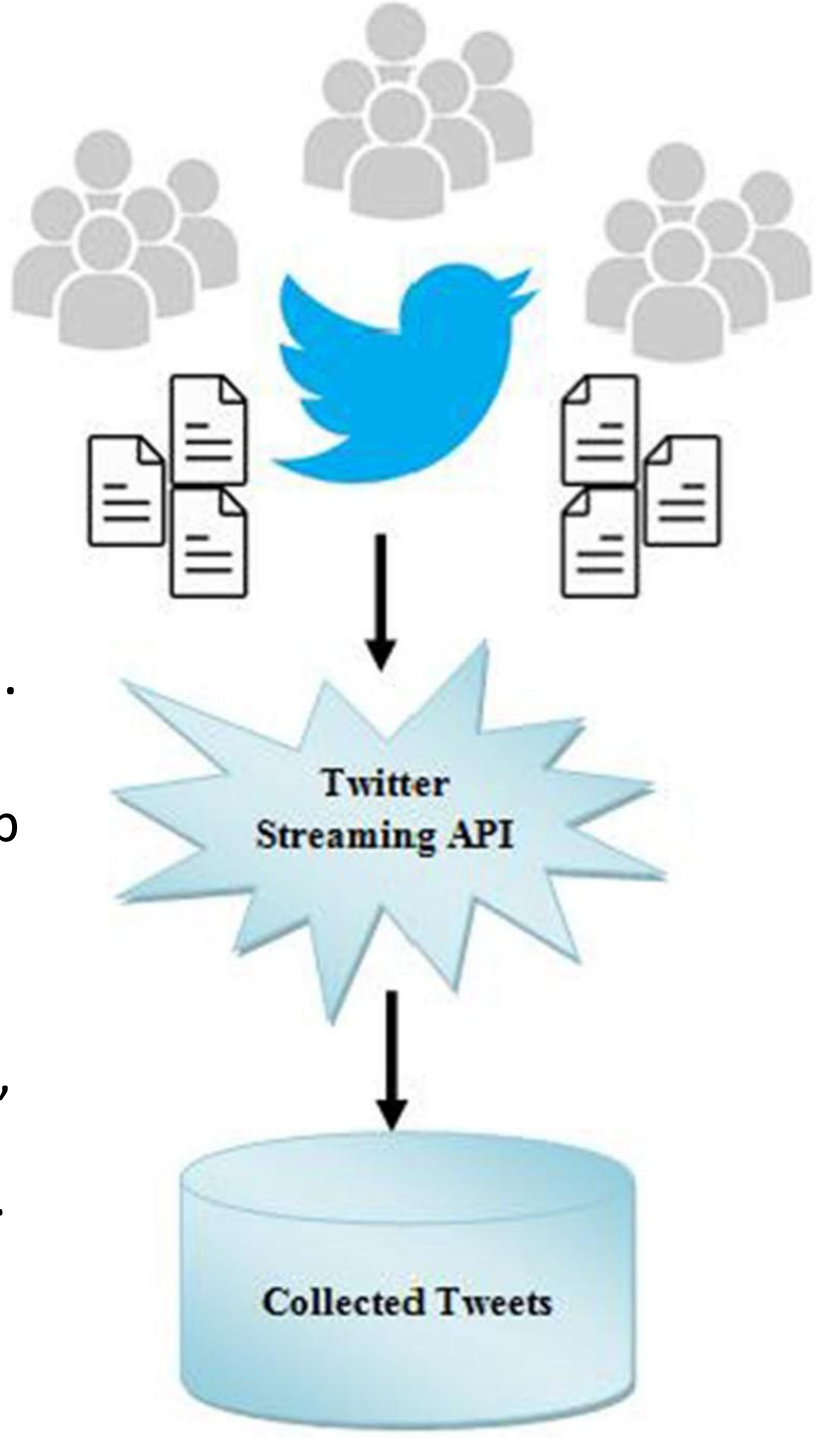
Data Collection through APIs

- Benefits of Using APIs for Data Collection:
  - **Structured Data:** APIs provide structured data in a consistent format, making it easier to extract and use.
  - **Real-Time Data:** Many APIs offer real-time or near-real-time access to the latest information.
  - **Efficiency:** APIs allow you to access specific data points without the need to scrape entire web pages.
  - **Reliability:** Data obtained through APIs is typically more reliable and accurate than web scraping.



Data Collection through APIs

- Example Scenario: Collecting Tweets Using the Twitter API:
  - API: Twitter API (requires an API key and authentication).
  - Steps:
    - Sign up for a Twitter Developer account and create an app to obtain API keys.
    - Use Python libraries like ***tweepy*** to send requests to the Twitter API endpoints.
    - Request tweets based on specific keywords, user profiles, or hashtags.
    - Parse the JSON response and extract relevant tweet data.
    - Store the data in a structured format for analysis.

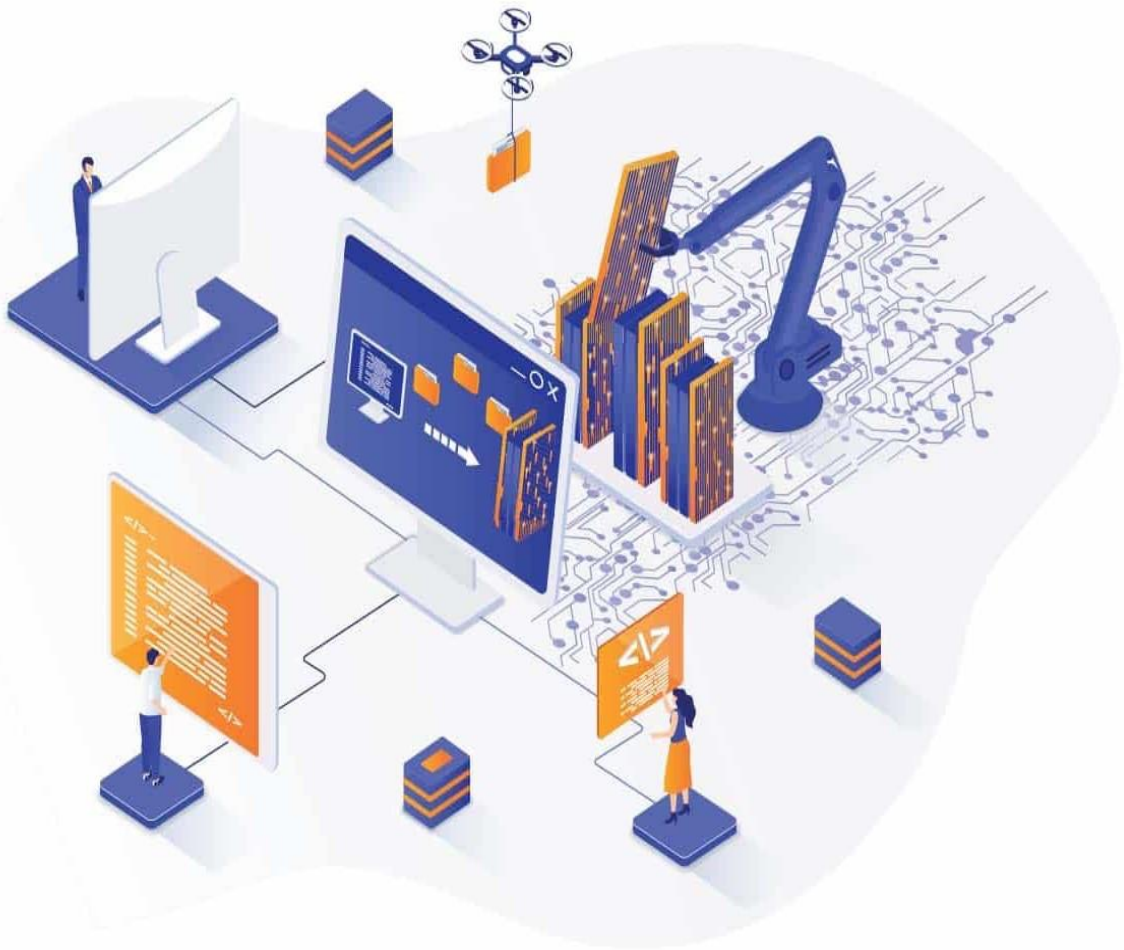


API: Best Practices

- **Rate Limiting**: Respect the API's rate limits to avoid being blocked.
- **Efficient Data Handling**: Parse and store data efficiently to manage large volumes.
- **Authentication**: Securely manage API keys and tokens.
- **Documentation**: Keep clear documentation of your API interactions for future reference.
- **Ethical Considerations**: Ensure compliance with data usage policies and respect user privacy.

API: Challenges

- **Rate Limits**: APIs often limit the number of requests that can be made in a given time period.
- **Data Quality**: Ensuring the accuracy and relevance of the data collected.
- **API Changes**: APIs may change over time, requiring updates to your data collection scripts.
- **Access Restrictions**: Some APIs have restricted access or require payment.



Data Collection by Web Scraping

1,097,398,145

Currently, there are around **1.1 billion** websites in the World. **18%** of these websites are active, **82%** are inactive.

193,527,411
websites are active

252,000
new websites are created **every day**

10,500
new websites are created **every hour**

175
new websites are created **every minute**

3
new websites are created **every second**

2,000+
new websites by the time you are done reading this article

Source: https://siteefy.com/how-many-websites-are-there/

Data Collection by Web Scraping

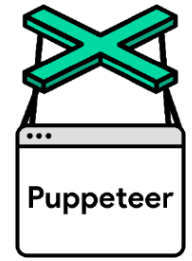
- Many valuable sources of text data are available online
  - news articles, social media content, discussion forums, etc.
- Web scraping enables us to gather relevant and up-to-date data for analysis and model training

Data Collection by Web Scraping

- Web Scraping Process:
  - HTTP Requests:
    - Web scraping starts with sending HTTP requests to a website's URL.
    - The website responds with HTML content, which contains the data we want to extract.
  - Parsing HTML:
    - The HTML content is parsed to extract specific elements (e.g., paragraphs, headings) containing the desired text data.
  - Data Extraction:
    - Use CSS selectors or XPath expressions to locate and extract the relevant data from the parsed HTML.
  - Data Transformation:
    - The extracted data might require further preprocessing, such as removing HTML tags, handling special characters, and tokenization.
  - Storing Data:
    - Store the extracted and preprocessed data in a structured format (e.g., CSV, JSON) for further analysis or model training.

Data Collection by Web Scraping

- Tools for Web Scraping
  - **Beautiful Soup:** A Python library for parsing HTML and XML documents. It provides easy methods to navigate and search the parsed content.
  - **Scrapy:** A more advanced Python framework for web scraping. It offers a structured approach to building web scrapers and handling complex websites.
  - **Requests:** A Python library for making HTTP requests, often used in conjunction with Beautiful Soup for simple scraping tasks.
  - **Selenium:** A Python library and tool used for automating web browsers to do a number of tasks including web-scraping to extract useful data and information.
  - **Puppeteer**: A Node.js library that provides a high-level API to control Chrome or Chromium.

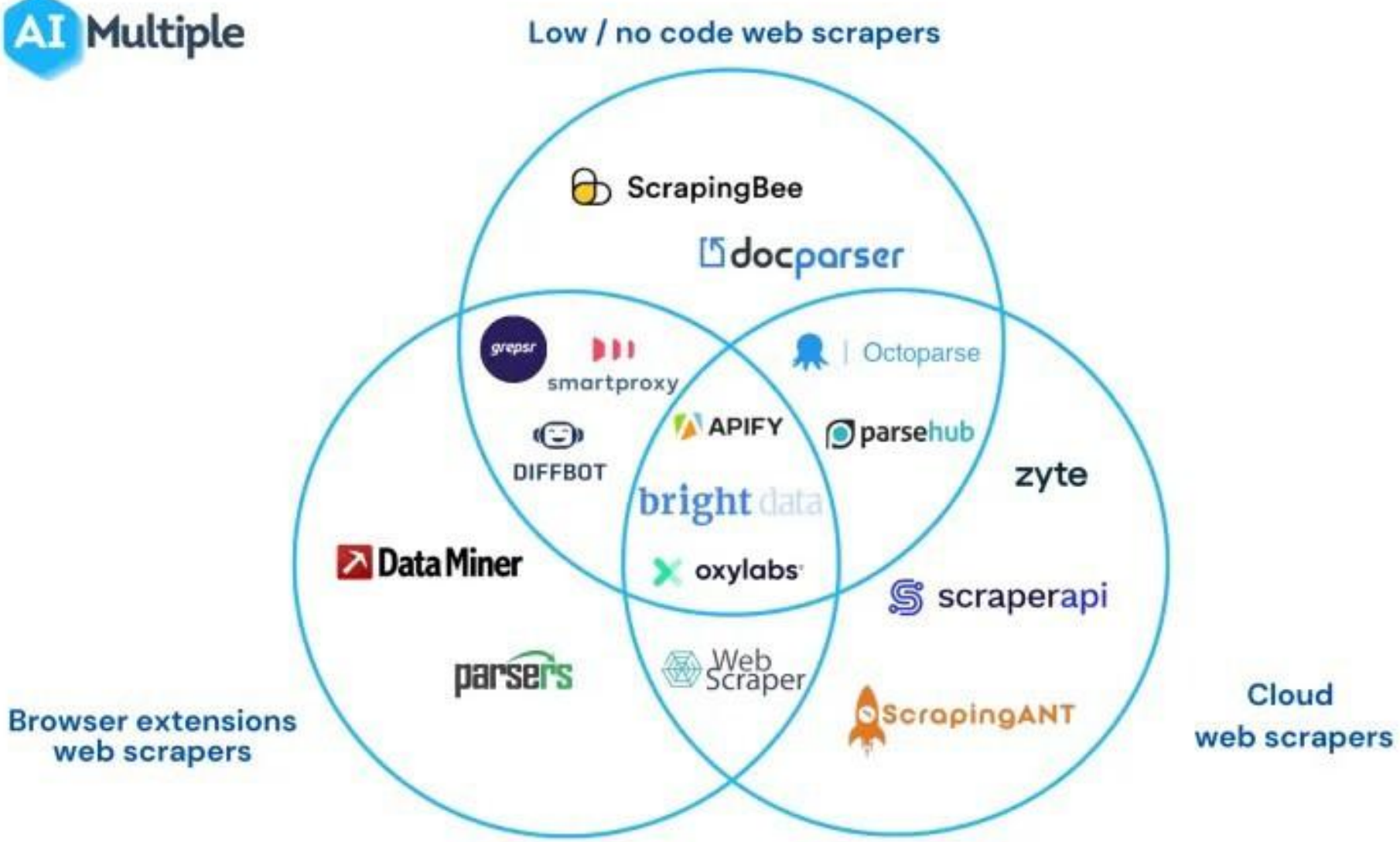


Data Collection by Web Scraping

- Tools for Web Scraping

	Beautiful Soup	Selenium	Scrapy
Performance	Slow for a few tasks	Faster than Beautiful Soup but has its limitations	Pretty fast and has the best performance out of the three
Extensibility	Best suited for small, low-complexity projects & beginners	Best suited for core JavaScript featured website	Best suited for large, complex project. Makes project robust & flexible
Ecosystem	Has a lot of dependencies in the ecosystem	Good ecosystem but can't use proxies	Can use proxies and VPMs and hence suitable for complex projects

AI Multiple



Source: https://research.aimultiple.com/web-scraping-tools/

Data Collection by Web Scraping

- Example Scenario: News Article Scraping

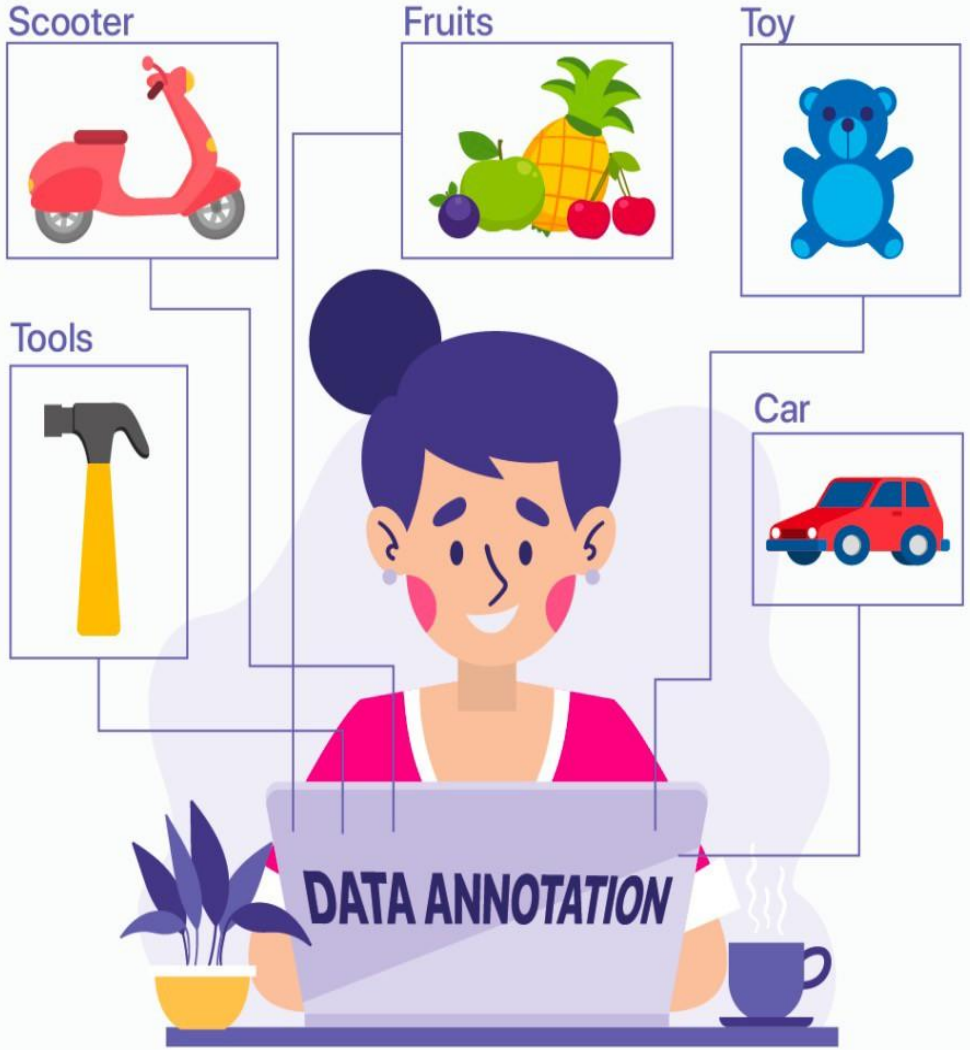
  - **Objective:** Gather recent news articles for a sentiment analysis project.
  - **Steps:**
    - Identify relevant news websites (e.g., major news outlets, specialized news blogs).
    - Send HTTP requests to article pages, retrieve HTML content.
    - Use BeautifulSoup to parse and extract article titles and content.
    - Preprocess extracted text (remove HTML tags, tokenize, etc.).
    - Store the preprocessed data for analysis.

Scraping: Best Practices

- **Respect robots.txt**: Check the website's robots.txt file to understand which parts of the site can be scrapped.
- **Rate Limiting**: Avoid overloading the server by limiting the rate of requests.
- **Use Proxies**: Distribute requests across multiple IP addresses to avoid getting blocked.
- **Error Handling**: Implement error handling to manage failed requests and retries.
- **Data Cleaning**: Ensure the scraped data is cleaned and structured appropriately.
- **Legal Compliance**: Ensure scraping is compliant with legal guidelines and the website's terms of service.
- **Privacy**: Be cautious with personal data to avoid violating privacy laws.

Scraping: Challenges

- **IP Blocking**: Websites may block IP addresses that send too many requests.
- **Captcha and Security Measures**: Some sites use Captchas and other techniques to prevent automated scraping.
- **Dynamic Content Loading**: Handling content that loads dynamically via JavaScript.
- …



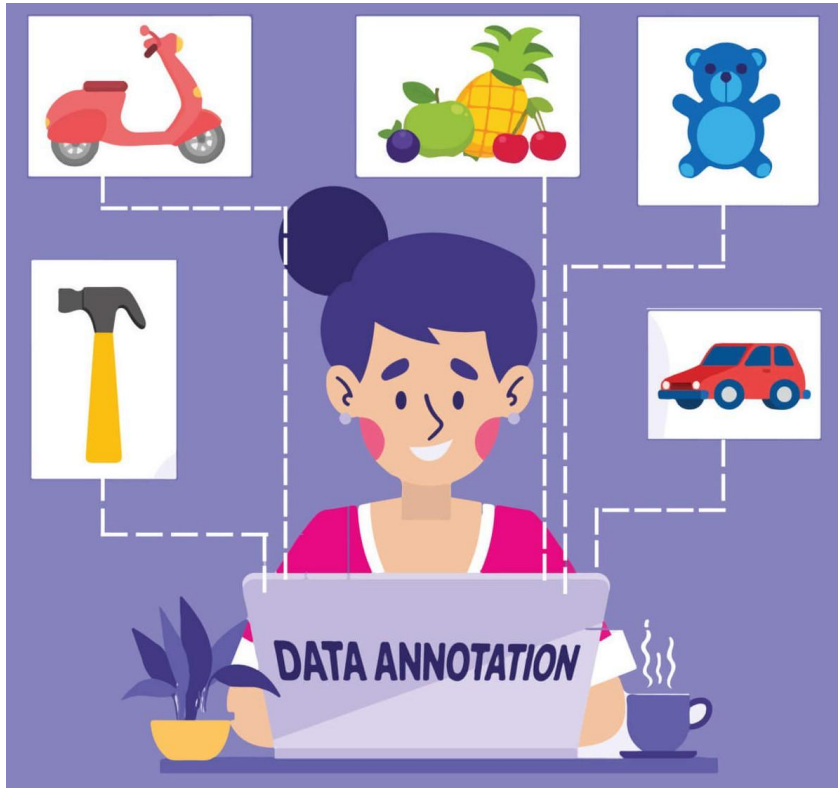
Data Annotation and Labeling

Data Annotation and Labeling

- Involve adding metadata or tags to raw text data, making it suitable for training supervised NLP models.
- High-quality annotations are crucial for building accurate and effective NLP systems

Importance of Annotation:

- Annotated data serves as ground truth for training and evaluating machine learning models.
- Properly labeled data contributes to the success of tasks like sentiment analysis, text classification, and named entity recognition.



Data Annotation and Labeling

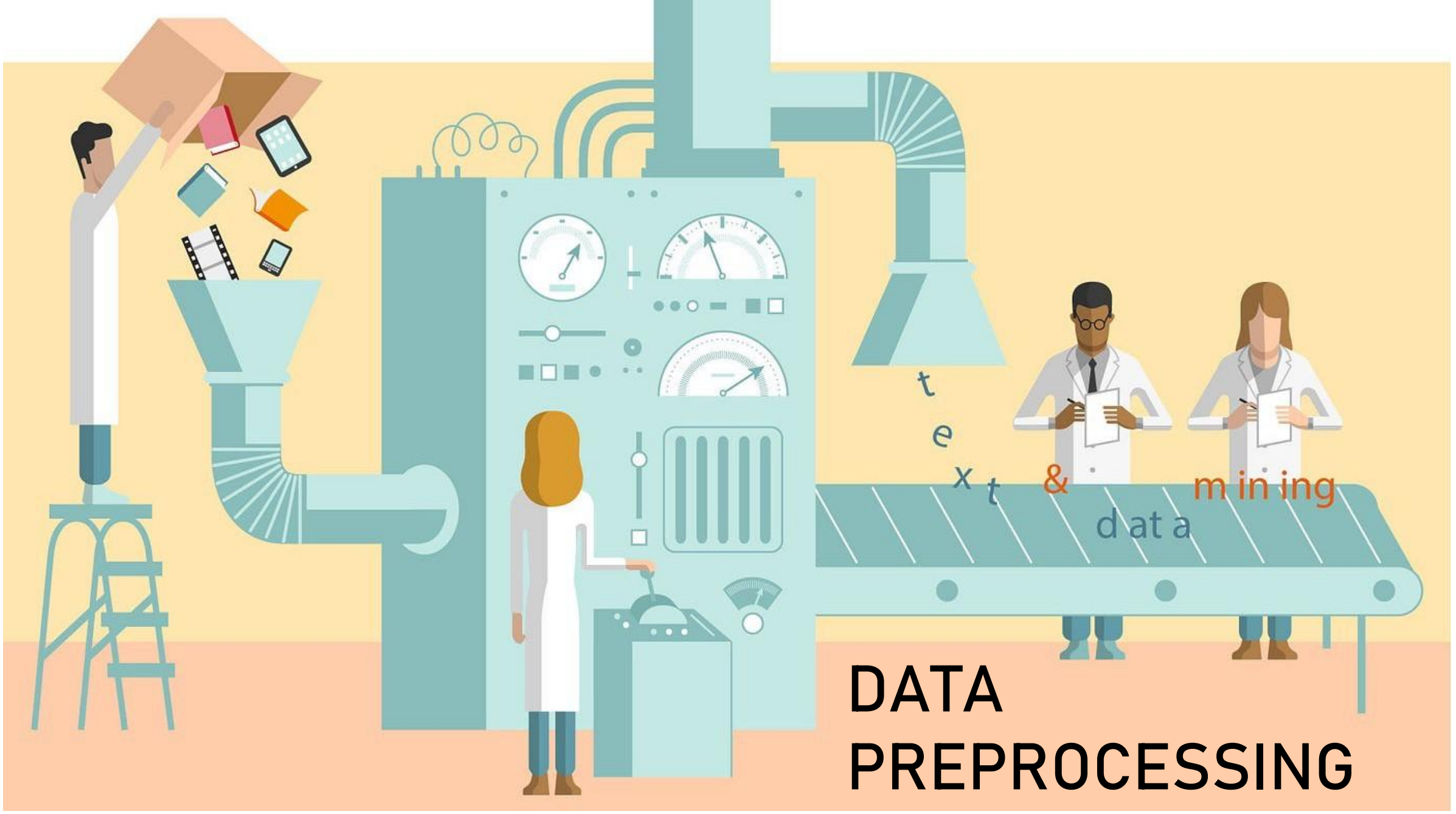
- Annotation Methods:
  - **Manual Annotation:**
    - Human annotators review and label each data point based on predefined guidelines.
    - Ensures high accuracy but can be time-consuming and expensive.
  - **Semi-Automatic Annotation:**
    - Annotators are aided by tools that suggest labels or generate annotations based on certain patterns.
    - Increases efficiency while maintaining human oversight.
  - **Crowdsourcing:**
    - Outsourcing annotation tasks to a large group of workers through platforms like Amazon Mechanical Turk.
    - Cost-effective for large datasets, but quality control is essential.

Data Annotation and Labeling

- Labeling Strategies:
  - **Single-Annotator:** One annotator labels each data point for consistency.
  - **Multiple-Annotator:** Multiple annotators label the same data point to assess inter-annotator agreement.
  - **Majority Voting:** Inconsistent labels are resolved by selecting the most common label among annotators.

Data Annotation and Labeling

- Challenges and Considerations:
  - **Ambiguity:** Some text may have multiple valid interpretations, leading to varying labels.
  - **Subjectivity:** Certain tasks, like sentiment analysis, can be subjective and result in differing annotations.
  - **Bias:** Annotator biases can impact the quality and fairness of labeled data.
  - **Continual Feedback:** Regular communication with annotators helps clarify guidelines and address questions.



DATA PREPROCESSING

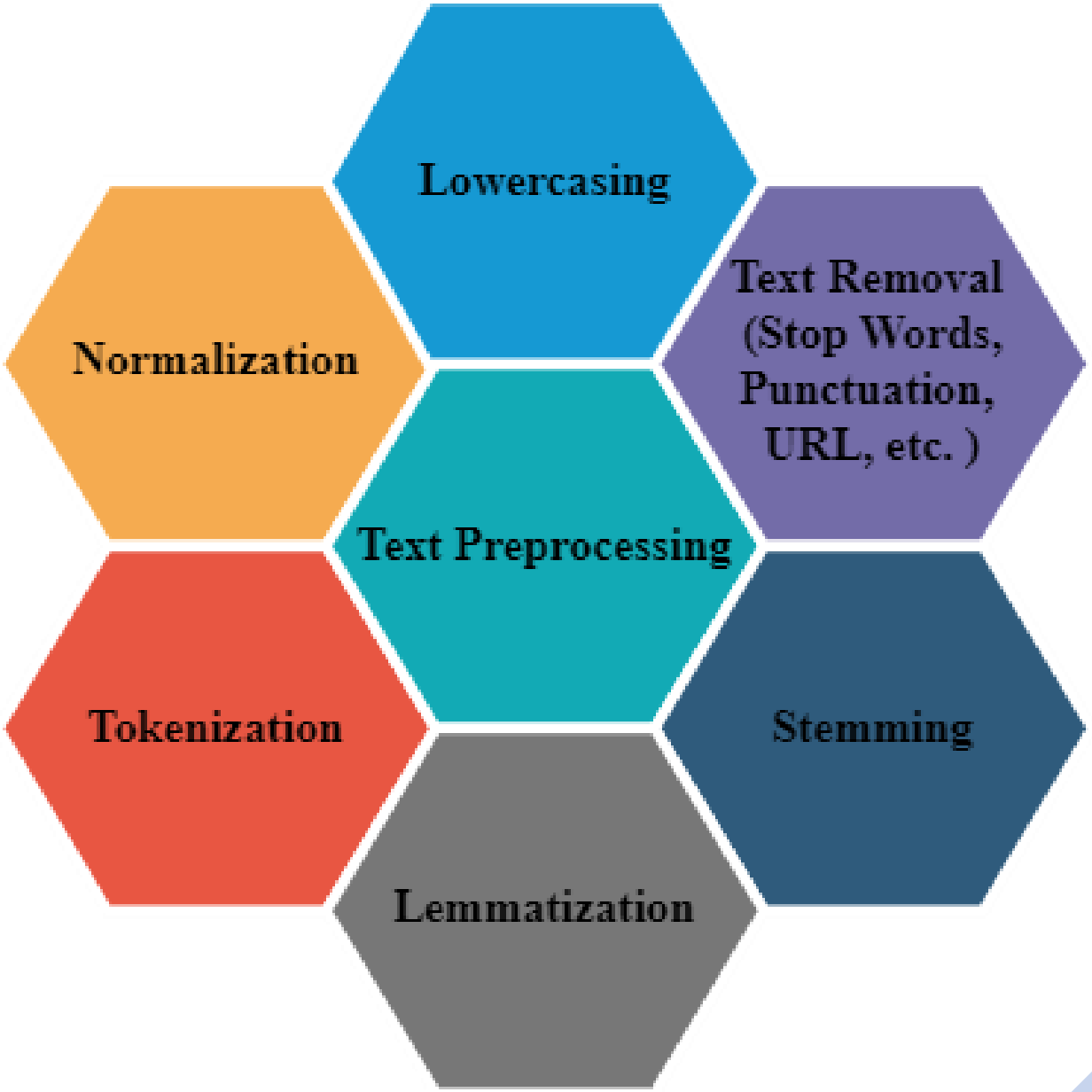
Data Preprocessing

- Definition

  Process of preparing and cleaning raw text data before it can be analyzed by natural language processing (NLP) algorithms or machine learning models.

- Goal

  Transform unstructured text data into a structured and standardized format that can be easily analyzed and understood by computers

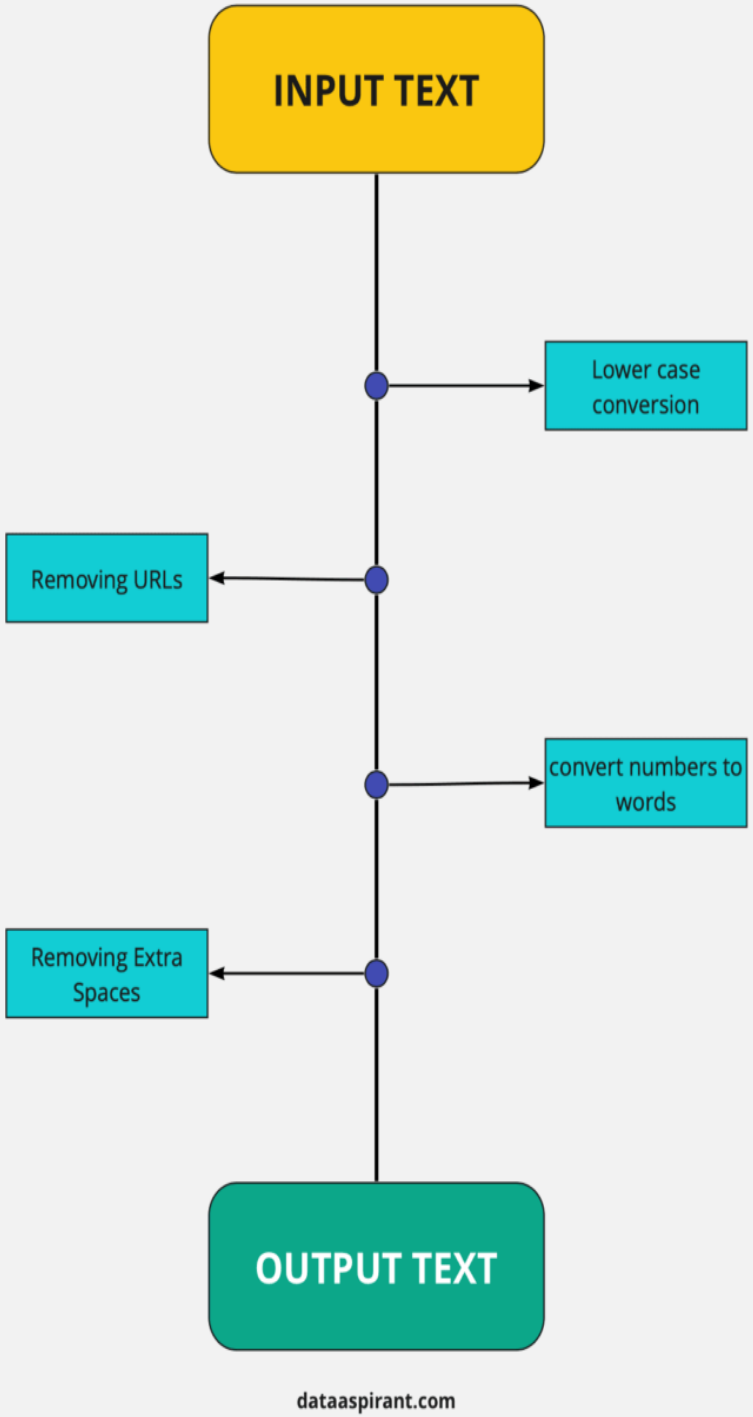

Lowercasing
Normalization
Text Removal (Stop Words, Punctuation, URL, etc.)
Text Preprocessing
Tokenization
Stemming
Lemmatization

Text Preprocessing

- Why Text Preprocessing is required?
  - Cleaning and standardizing the data
  - Reducing the vocabulary size
  - Enabling efficient text representation
  - Improving model accuracy
  - Facilitating text understanding
  - …

Text Preprocessing

- Types of text preprocessing techniques
  - Tokenization
  - Lowercasing
  - Stop word removal
  - Stemming and Lemmatization
  - Removing special characters and digits
  - Removing URLs and email addresses
  - Spell checking and correction
  - …



Tokenization

- Process of breaking up a text into individual words, phrases, symbols or other meaningful elements, called *tokens*
- Tokens are used as the basic building blocks
- Involves identifying the boundaries between words or other meaningful elements in a text
- A text can be split on whitespace or punctuation marks such as spaces, commas, periods, question marks, and exclamation marks

Tokenization

- For example, consider the following sentence:

  "John loves playing soccer with his friends.“

- The tokens in this sentence after tokenization might be:
  - John
  - loves
  - Playing
  - soccer
  - with
  - his
  - friends

Lowercasing

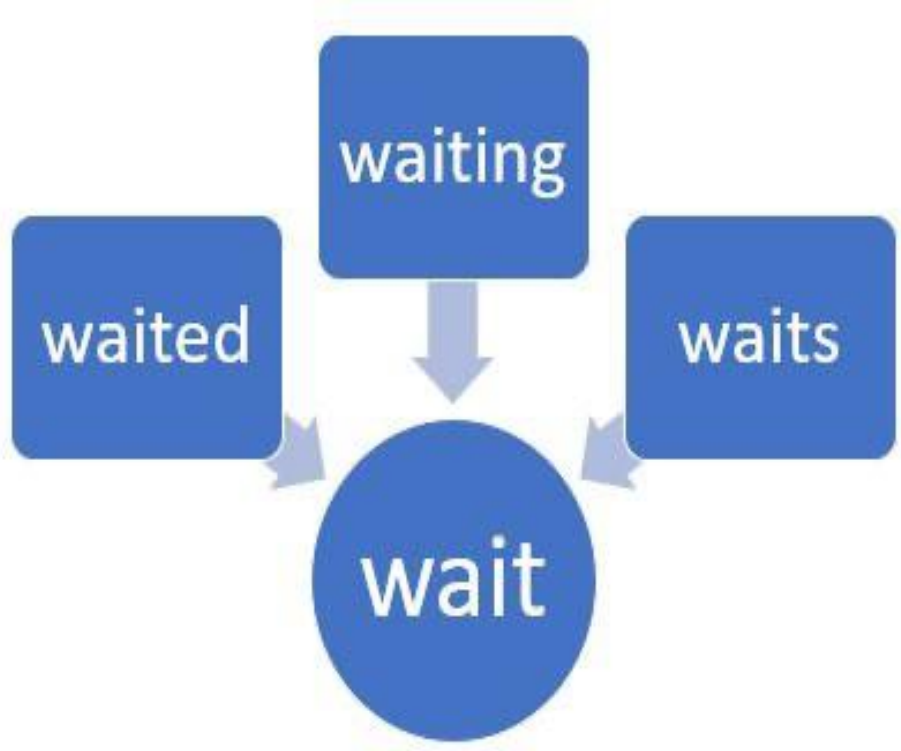
- Lowercasing ALL your text data
- One of the simplest and most effective form of text preprocessing
- Applicable to most text mining and NLP problems
- Useful to ensure that the same word is not treated differently because of its capitalization
- Help in cases where your dataset is not very large
- Significantly helps with consistency of expected output

Lowercasing

Raw	Lowercased
Canada CanadA CANADA	canada
TOMCAT Tomcat toMcat	tomcat

Stemming

- Reduce inflected or derived words to their base form, known as the stem
- Stem may or may not be meaningful
- Purpose is to simplify the text data and reduce the vocabulary size
- Make easier to process and analyze
- Involves applying a set of rules or algorithms to the words in the text data to identify and remove affixes (prefixes and suffixes)
- Porter stemming, Snowball stemming, Lancaster stemming



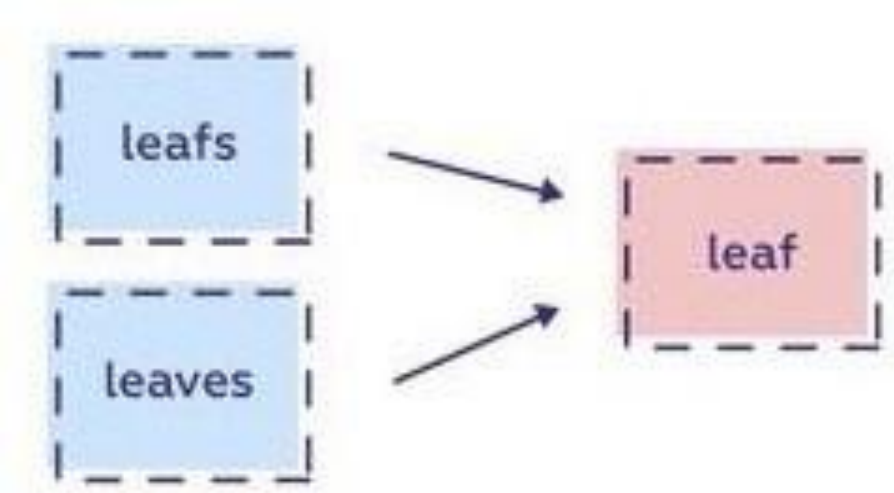
Stemming

	original_word	stemmed_words
0	connect	connect
1	connected	connect
2	connection	connect
3	connections	connect
4	connects	connect

	original_word	stemmed_word
0	trouble	troubl
1	troubled	troubl
2	troubles	troubl
3	troublemsome	troublemsom

Lemmatization

- Very similar to stemming
- Reduce inflected or derived words to their base or dictionary form, known as the lemma
- Lemma should have a meaning
- Considers the context and part of speech of the word in order to produce the correct lemma
- Involves the use of a lexicon or a set of rules
- More accurate technique than stemming
- More computationally expensive and slower than stemming

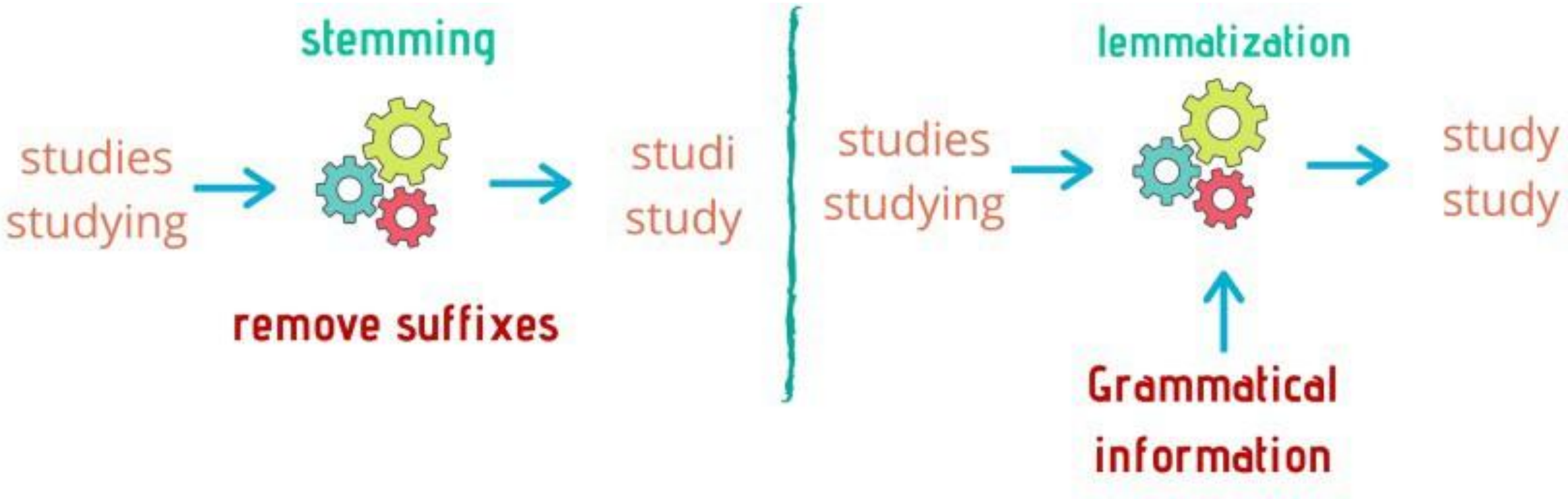


Lemmatization

	original_word	lemmatized_word
0	trouble	trouble
1	troubling	trouble
2	troubled	trouble
3	troubles	trouble

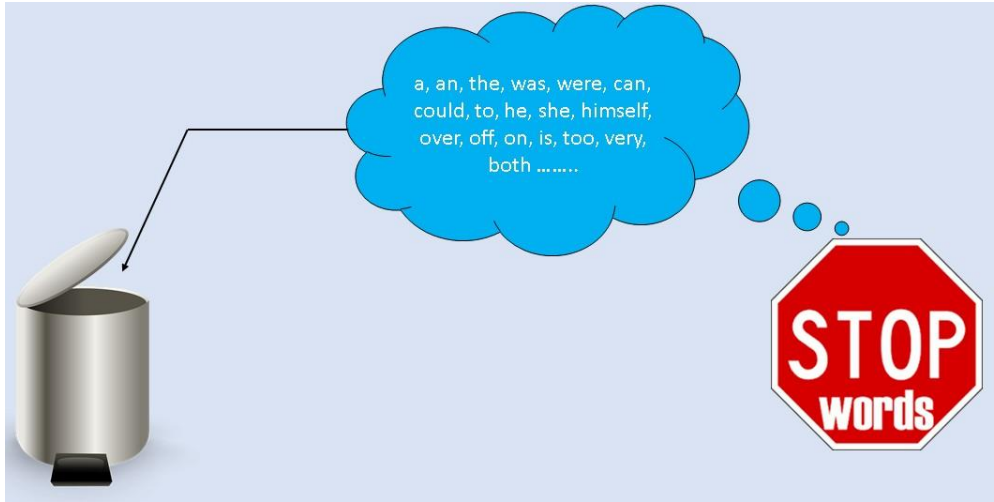
	original_word	lemmatized_word
0	goose	goose
1	geese	goose

Stemming vs Lemmatization



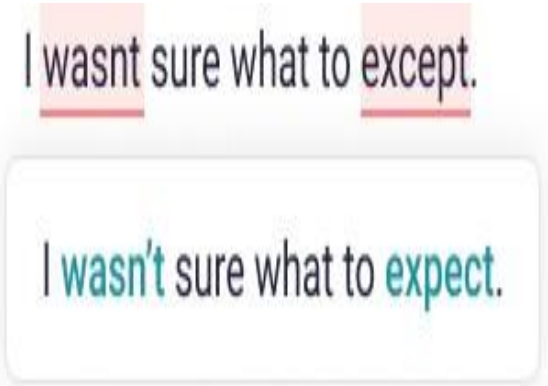
Stop words Removal

- Remove words that are considered irrelevant or redundant in the context of the text data being analyzed
- Stop words are common words that occur frequently in language
- "a", "an", "the", "and", "in", "on", and "at“, etc. - do not carry much meaning on their own and can be safely ignored in many text analysis tasks
- Purpose is to reduce the size of the vocabulary and simplify the text data
- Can be easily implemented using pre-built lists of stop words or by creating custom lists for specific text analysis tasks



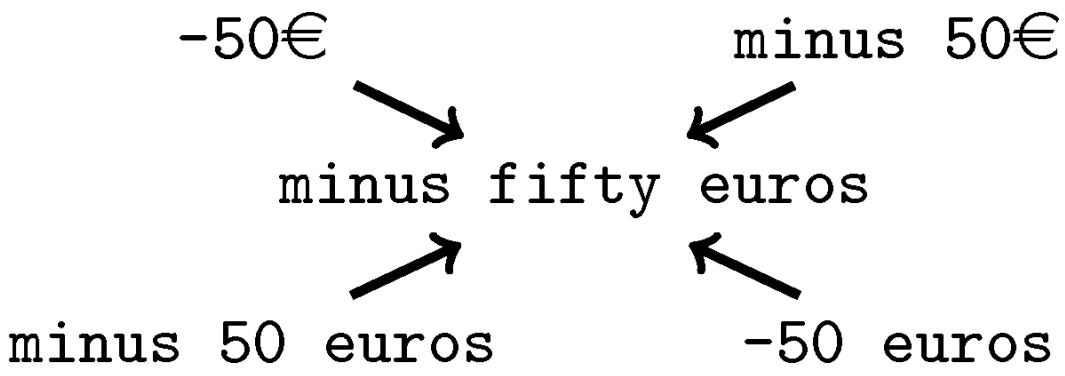
Spell Checking & Correction

- Identify and correct spelling errors in the text data
- Spelling errors can occur due to typos, misspellings, or other mistakes made during the text entry or transcription process - can affect the accuracy and quality of the text analysis results
- Dictionary-based spell checking
- Rule-based spell checking
- Statistical spell checking
- Machine learning-based spell checking
- Mainly useful in applications that require precise language processing, such as sentiment analysis, named entity recognition, and machine translation



Text Normalization

- Transform text data into a standardized format that is easier to process and analyze
- Purpose is to reduce the variability in text data and increase its consistency and accuracy
- Case normalization
- Punctuation removal
- Number normalization
- Symbol and abbreviation expansion
- Accent and diacritic removal …

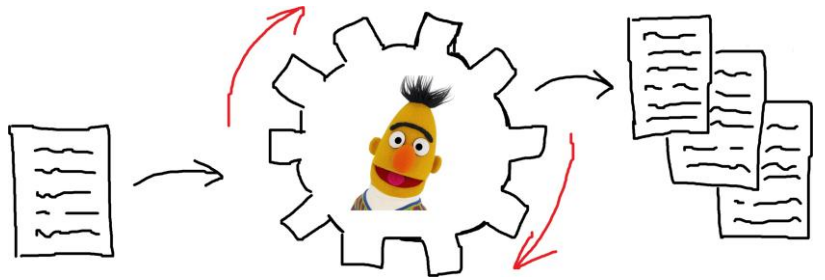


Text Normalization

Raw	Normalized
2moro 2mrrw 2morrow 2mrw tomrw	tomorrow
b4	before
otw	on the way
:) :-) ;-)	smile

Text Enrichment / Augmentation

- Enhance the quality and quantity of text data by adding or generating new content
- Purpose is to increase the diversity and relevance of the text data, and provide additional context and information for text analysis tasks
- **Part-of-speech tagging**: involves labeling each word in the text data with its corresponding part of speech, such as noun, verb, or adjective
- **Named entity recognition**: involves identifying and labeling named entities in the text data, such as people, organizations, and locations
- **Text generation**: involves generating new text data based on the existing text data, using techniques such as language models, recurrent neural networks, and Markov chains



Do you need it all?

- **Must Do:**
  - Noise removal
  - Lowercasing (can be task dependent in some cases)
- **Should Do:**
  - Simple normalization — (e.g. standardize near identical words)
- **Task Dependent:**
  - Advanced normalization (e.g. addressing out-of-vocabulary words)
  - Stop-word removal
  - Stemming / lemmatization
  - Text enrichment / augmentation

General Rule of Thumb

- Not all tasks need the same level of preprocessing

Level of Text Preprocessing Needed

	Domain Specific / Noisy Texts	General / Well Written Texts
Lots of data	- Moderate pre-processing - Text enrichment could be helpful	- Light pre-processing - Text enrichment could be helpful, but not critical
Sparse data	- Heavy pre-processing - Text enrichment is important	- Moderate pre-processing - Text enrichment could be helpful

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